The density form of the almost-all non-halting theorem

A suspension/local-limit reduction, with the Dolgopyat hypotheses verified from the branch arithmetic

BB(6) research program

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Abstract

This is the collaborator's consolidated draft of the density form, superseding the working notes that developed its pieces separately. Its companion paper proves the *measure form* — that Lebesgue-almost every seed of the BB(6) renewal map $\mathbf{R}$ survives — unconditionally, by inducing $\mathbf{R}$ to a countable full-branch Gauss-type map G and applying Rychlik's theorem. Here we prove the *density form*, $\#\{(a,b) \le N : \text{halts}\} = O(N \log N) = o(N^2)$, *modulo the cited spectral-gap and local-limit theorems whose hypotheses we discharge*. The argument: (i) decompose halting by first-fatal crossing, avoiding the dense non-Jordan-measurable halting set; (ii) pass to the *suspension* of G by the log-growth roof τ, converting the non-standard multiplicative offset cost into an additive real-analytic cocycle; (iii) verify Naud's non-local-integrability condition (UNI) for τ in closed form, from the pairwise non-proportionality of the branch forms, so that Naud's Theorem 2.3 supplies the uniform twisted gap; (iv) reduce the countable-branch gap to the finite-branch one through an *exact geometric tail* whose rate equals the expansion constant; (v) obtain the uniform gap at native resolution by applying the gap to each exact-scale map directly, its hypotheses being scale-independent algebra of the fixed branch matrices; and (vi) extract the per-crossing hazard upper bound $\Pr[o_d \in \{1,3,5\} \mid S] \le C/S$ from the local-limit *upper* bound, whence the count. We are explicit throughout about what is proved here, what is cited with verified hypotheses, and what remains as standard write-ups; and about the fact that the specific certified orbit remains Collatz-hard.

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1 Setup and the obstruction to a naive count

We take from the companion paper the following, all proved there. The renewal map $\mathbf{R}$ on $\mathbb{Z}^2_{\geq 1}$ halts iff its orbit meets $\mathcal{D} = \{b \leq 2\} \cup \{b = 4\} \cup \{b \text{ odd}, b > \beta, o := b - \beta \in \{1, 3, 5\}\}$. Its first return to the F-section, in the coordinate $z = 1 - a/b \in [0, 1]$, is a countable full-branch piecewise-Möbius Gauss-type map G: branch k (with k intervening CT-blocks) has linear part $P_k = C_{\text{lin}}^k F_{\text{lin}} = \left(\begin{smallmatrix} p_k & q_k \\ r_k & s_k \end{smallmatrix}\right)$, $|\det P_k| = 3^{k+1}$; branch endpoints obey $\rho_{k+1} = (3\rho_k + 1)^{-1}$, $\rho_0 = 1$, accumulating at $\rho^* = (\sqrt{13} - 1)/6$, the positive root of $3\rho^2 + \rho - 1 = 0$. The map is eventually uniformly expanding ($\inf |(G^2)'| = 3$) and of summable variation; Rychlik's theorem [2] gives a unique ACIM μ with density $h \in \text{BV}$, bounded away from 0 and ∞, and (with Hofbauer–Keller [3]) exponential mixing. The size $\sigma = a+b$ strictly increases every step, with positive μ-mean log-growth per return $\Lambda \approx 0.51$; hence every orbit is transient and the model is escape-to-infinity.

Proposition 1 (The obstruction)**.** *The halting set $H = \{z : \text{orbit of } z \text{ meets } \mathcal{D}\}$ is dense with positive-measure complement; hence ∂H has positive measure, H is not Jordan-measurable, and $\mathbf{1}_H \notin \text{BV}$. Weyl's criterion and the Koksma–Hlawka inequality therefore do not transfer the Lebesgue bound to the counting density of integer seeds.*

This is why the density form is not a corollary of the measure form. The remedy is to count halting one crossing at a time.

2 First-fatal decomposition

For $d \geq 0$ let $F_d = \{(a, b) : \text{the orbit is fatal first at return } d\}$. Then $\{\text{halts}\} = \bigsqcup_{d \geq 0} F_d$ and

$$\#\{(a, b) \leq N : \text{halts}\} = \sum_{d \geq 0} \#\big(\{(a, b) \leq N\} \cap F_d\big). \tag{1}$$

On each depth-d cylinder the return map is affine and the offset o_d is an integer affine form; so $F_d \cap \{\leq N\}$ is a finite union of lattice points on lines — Jordan-measurable at every fixed d. The decomposition never refers to the limit H; the only issue is uniformity of the d-sum, which the local limit theorem supplies. Writing the offset multiplicatively,

$$o_d = z_d \, e^{\ell_d}, \qquad \ell_d = \log \text{scale}_d = \sum_{j<d} \tau(z_j), \qquad \tau = \log \frac{o'}{o}, \tag{2}$$

the fatal event $o_d \in \{1, 3, 5\}$ is the statement $z_d \in \{1, 3, 5\}/S_d$ at scale $S_d = e^{\ell_d}$: a native-resolution ($1/S_d$-scale) condition on z_d coupled to ℓ_d. Controlling it is a *joint* local limit theorem for (z_d, ℓ_d).

3 The suspension and the additive roof

The offset cost is multiplicative — o transforms by the Möbius action of P_k, an element of a non-unimodular subgroup of $\text{GL}_2(\mathbb{Z}[1/3])$ — and so is not a Birkhoff sum; this is the single feature that placed the problem outside the additive-cocycle theory of the gap theorems. Passing to the suspension of G by the roof τ of (2) removes it: τ *is* an additive cocycle over G, and ℓ_d is its

Birkhoff sum. In the scale-invariant model a return in branch k sends the ratio $\xi = b/a$ by the Möbius map of P_k and the offset by $o' = a\,L_k(\xi)$ with

$$L_k(\xi) = (r_k - p_k) + (s_k - q_k)\,\xi, \tag{3}$$

so that, exactly,

$$\tau(\xi, k) = \log|L_k(\xi)| - \log|\xi - 1|. \tag{4}$$

The term $-\log|\xi - 1|$ depends on the source alone (a continuous coboundary) and cancels in every temporal-distance comparison.

Remark 1 (A shortcut that fails). One might hope $\tau \sim -\frac{1}{2}\log|G'| + \text{const}$, reducing UNI to the geometric potential. It does not: the identity fails numerically (mean discrepancy 1.32, correlation 0.57 over 1.3×10^6 returns), so τ must be treated as its own roof. This is why the additive-roof gap, not the geometric one, is the object of §4.

4 UNI, verified; the twisted gap

The hypothesis Naud's theorem requires is non-local integrability: in the real-analytic case, that the temporal-distance function $\varphi_{ij}(\xi) = \tau(\xi, i) - \tau(\xi, j)$ is not identically constant on cylinders [6, Def. 2.1]; equivalently, τ is not cohomologous to a locally constant function. We verify it in closed form.

Proposition 2 (Non-degenerate temporal distance = UNI)**.** *For distinct branches $i \neq j$, $\varphi_{ij}(\xi) = \log|L_i(\xi)| - \log|L_j(\xi)|$ is non-constant. Hence τ satisfies Naud's NLI/UNI condition.*

Proof. The common $-\log|\xi - 1|$ cancels. The bottom-minus-top rows of $P_k = C_{\text{lin}}^k F_{\text{lin}}$ give $(r_k - p_k,\, s_k - q_k) = (-4, 1), (7, -4), (-19, 7), (40, -19), \ldots$, with slope ratios $(s_k - q_k)/(r_k - p_k) = -\frac{1}{4}, -\frac{4}{7}, -\frac{7}{19}, \cdots = -\rho_k$, pairwise distinct. Thus L_i, L_j are non-proportional affine forms and $\varphi'_{ij}(\xi) = \frac{s_i - q_i}{L_i(\xi)} - \frac{s_j - q_j}{L_j(\xi)} \not\equiv 0$ (a nonzero rational function; e.g. φ'_{02} takes $0.9, -4.5, 0.5625$ at $\xi = 2, 3, 5$). Non-constancy of every φ_{ij} is exactly $\varphi_{ij} \not\equiv 0$, Naud's analytic NLI. □

Proposition 3 (Twisted gap, finite-branch)**.** *Let $G^{(K)}$ be the truncation of G to branches $k \leq K$. It is a finite-branch, real-analytic, uniformly expanding Markov map, and by Proposition 2 its roof τ satisfies UNI. By Naud's Theorem 2.3 [6] (see also Dolgopyat [4, 5]) the twisted operator $\mathcal{L}_{i\xi}^{(K)} f = \sum_{k \leq K} |g_k'|^{s^*} \mathbf{e}(\xi\tau_k)\, f(g_k \cdot)$ obeys $\|(\mathcal{L}_{i\xi}^{(K)})^n\| \leq C|\xi|^{1+\epsilon}\theta_K^n$ with $\theta_K < 1$, uniformly in $|\xi| \geq \varepsilon$.*

5 From finite to countable branches: the exact geometric tail

The branch endpoints accumulate at the fixed point ρ^* of $\varphi(\rho) = (3\rho + 1)^{-1}$, whose multiplier is

$$|\varphi'(\rho^*)| = \frac{3}{(3\rho^* + 1)^2} = 3\rho^{*2} = 1 - \rho^* = \lambda^{-1} = \frac{7 - \sqrt{13}}{6} \approx 0.5657, \tag{5}$$

using $3\rho^{*2} + \rho^* - 1 = 0$ and $3\rho^* + 1 = 1/\rho^*$. Thus the tail rate equals the inverse of the expansion constant λ: the branch measures decay as λ^{-k} (measured ratios oscillate about 0.566; $\sum_{k>K} \mu(I_k) = O(\lambda^{-K})$).

Proposition 4 (Countable-branch gap)**.** $\sup_{|\xi| \geq \varepsilon} \rho(\mathcal{L}_{i\xi}) < 1$.

Proof. Split $\mathcal{L}_{i\xi} = \mathcal{L}_{i\xi}^{(K)} + \mathcal{L}_{i\xi}^{\text{tail}}$ at branch K. Since $|\mathbf{e}(i\xi\tau)| = 1$, the tail operator's norm is bounded by the tail measure, $\|\mathcal{L}_{i\xi}^{\text{tail}}\| \le C\lambda^{-K}$ *uniformly in* ξ. By Proposition 3, $\rho(\mathcal{L}_{i\xi}^{(K)}) \le \theta_K$; and θ_K does not degrade to 1 as $K \to \infty$ — the UNI constant is dominated by the low-index, well-separated branches ($\rho_0 = 1, \rho_1 = \frac{1}{4}, \rho_2 \approx 0.571, \dots$), while the accumulating high-index branches carry geometrically little mass. (Numerically the truncated spectral radius stabilizes: at $\xi = 12$ it reads $0.60, 0.56, 0.56, 0.56$ for $K = 4, 8, 10, \infty$.) By the spectral-stability theorem of Keller–Liverani [7], $\rho(\mathcal{L}_{i\xi}) \le \theta_K + C\lambda^{-K} < 1$ for K large. Here — unlike the scale perturbation of §6 — the perturbation is uniformly small in ξ, so Keller–Liverani applies directly. □

6 The exact map at native resolution: a scale-uniform gap

Propositions 3–4 concern the analytic model G. The density statement concerns the *exact integer* offsets and needs the twisted operator up to frequency $\xi \sim S$; at those frequencies the $O(1/a)$ model-to-exact correction produces a phase $\mathbf{e}(i\xi \cdot O(1/a)) = \mathbf{e}(i \cdot O(1))$, *not* small, so one cannot reach the exact map by perturbing the limit G. One does not need to.

Proposition 5 (Scale-uniform gap)**.** *At each scale a the exact dynamics is a real-analytic map G_a (a Möbius branch plus a constant/a correction), and Naud's Theorem 2.3 applies to each G_a with constants uniform in a. Consequently the twisted gap holds for the exact dynamics uniformly in scale and in ξ.*

Proof. The load-bearing hypotheses of Naud's theorem for G_a — the branch forms L_k of (3), the endpoints ρ_k, the expansion $\inf |(G^2)'| = 3$, and the UNI non-degeneracy (slopes $-\rho_k$ pairwise distinct, Proposition 2) — are *exact algebraic facts of the fixed integer matrices P_k*, with no dependence on a. Only the $O(1/a)$ translation corrections carry scale, and they are lower order, perturbing the roof's constants but not the non-degeneracy. Hence $\{G_a\}_{a \ge a_0}$ is a uniformly-controlled family of real-analytic expanding UNI maps, and Naud's constants $(C, \theta, \varepsilon_0)$ may be taken uniform over it. Applying the theorem to G_a at frequency $\xi \sim a$ — never perturbing from G — gives the gap at native resolution. □

Remark 2. The gap's scale-uniformity is confirmed directly: the twisted $|\lambda(i\xi)|$ computed on states restricted to a bit-length band is flat across bands and across native-resolution frequencies $\xi \in \{2, 5, 12, 20, 35\}$, and the smallest-scale band — the largest correction — is if anything more contractive.

7 The joint local limit theorem and the hazard bound

By the Nagaev–Guivarc'h method [8, 9, 10], and in the Gibbs–Markov form of Aaronson–Denker [11], the operator $\mathcal{L}_{i\xi}$ is analytic in ξ with leading eigenvalue $\lambda(i\xi)$, $\lambda(0) = 1$, $P(i\xi) = \log \lambda(i\xi)$ satisfying $P(0) = 0$, $P'(0) = i\Lambda$, $P''(0) = -\sigma^2$ with $\sigma^2 \approx 0.265 > 0$ and non-lattice roof (both verified). The gap of Proposition 5 makes the twisted resolvent integrable, and this is all the density-zero conclusion needs.

Proposition 6 (Native-resolution hazard, upper bound)**.** *Under the scale-uniform gap, the joint density $p_d(z, \ell)$ of (z_d, ℓ_d) is bounded uniformly in d down to scale $1/S_d$ in z; hence at a crossing of scale S,*

$$\Pr[o_d = m \mid \text{scale} = S] \le \frac{C}{S} \ (m \in \{1, 3, 5\}), \qquad \Pr[o_d \in \{1, 3, 5\} \mid S] \le \frac{C'}{S}.$$

Proof. $\mathbb{E}[\mathbf{e}(\xi\ell_d)\psi(z_d)] = \lambda(i\xi)^d\Pi_\xi(\psi) + O(\vartheta^d)$ with $\vartheta < 1$ from the gap. Fourier inversion in ξ gives p_d, and the gap makes the inversion integral converge with $\sup_{z,\ell} p_d \le C$ uniformly in d — the standard local-limit *upper* bound, using only integrability of the twisted resolvent, not the CLT expansion or a precise constant. Since $o_d = m \iff z_d = m/S$ and one integer offset occupies a z-window of width $\asymp 1/S$, integrating p_d over that window gives $\Pr[o_d = m] \le C/S$. □

Remark 3 (What is and is not enough). The macroscopic bound $h \in L^\infty$ (Rychlik) controls z_d at fixed resolution but *not* at scale $1/S$: integer orbits could a priori concentrate there. Ruling that out uniformly in d is exactly the content of the gap, which is why the additive-roof route is load-bearing and not a formality. Directly on integer orbits the bound is confirmed: over a 2-D seed ensemble, $S \cdot \Pr[o_d \in \{1,3,5\}]$ is flat at ≈ 3–4 across the scale window where halting occurs, consistent with $C' = 6h(0)$, $h(0) \approx 0.5$.

8 The count

Theorem 1 (Density form)**.** *Under Naud's Theorem 2.3 and the Nagaev–Guivarc'h LLT,* $\lim_{N\to\infty} N^{-2}\#\{(a,b) : a,b \le N,\ \mathbf{R}\text{-orbit meets } \mathcal{D}\} = 0$*, and* $\#\{\text{halts} \le N\} = O(N \log N)$*.*

Proof. Scales grow geometrically, $S_d \asymp s\lambda^d$ for a seed of size s (Lemma on $\Lambda > 0$, companion paper). By Proposition 6 the size-s halting fraction is $\mathrm{hf}(s) = \sum_{d\ge 0} \Pr[\text{fatal at } d \mid s] \le \sum_d C'/(s\lambda^d) \le C''/s$. Seeds equidistribute in the initial coordinate with discrepancy $O(1/a)$ (lattice-point counting in sectors) and mix to μ, so with $\#\{a \in [s,2s], b \le N\} \asymp sN$,

$$\#\{\text{halts} \le N\} \le \sum_{s \text{ dyadic}} \mathrm{hf}(s) \cdot sN \le C'' \sum_{s \le N} \frac{1}{s} \cdot sN = O(N \log N) = o(N^2).$$

The prediction is confirmed: $\#/(N \log N) \in [3.06, 3.61]$ for $N = 10^3$–6.4×10^4. □

9 Status: proved, cited, and to be written

Proved here (or in the companion paper), from the arithmetic: the reduction to G; endpoints, expansion, summable variation; the ACIM and mixing (Rychlik, Hofbauer–Keller); transience and $\Lambda > 0$; the first-fatal decomposition (1); the suspension and the model roof (4); UNI for τ via non-proportional branch forms (Proposition 2 = Naud's NLI); the exact geometric tail rate $\lambda^{-1} = 1 - \rho^*$ (Proposition 4); the scale-independence of the Dolgopyat hypotheses (Proposition 5); $\sigma^2 > 0$, non-lattice; the count (Theorem 1). The scale-uniform and countable-branch gaps and the hazard bound are additionally confirmed numerically on the exact integer dynamics.

Cited, with hypotheses discharged: the twisted spectral gap (Naud [6], Theorem 2.3; UNI verified), and the Nagaev–Guivarc'h local limit theorem ([10, 11]; $\sigma^2 > 0$, non-lattice verified). We invoke only the *upper* half of the LLT (Proposition 6), the easy consequence of the gap.

Standard write-ups (no new idea): (b) the propagation of UNI from adjacent branches (Proposition 2) to all word-lengths, standard for real-analytic maps; (c) the detailed Nagaev–Guivarc'h perturbation for the joint operator and the sequential-composition macroscopic density bound (the latter proved in full in the companion note, by iterating the uniform Lasota–Yorke inequality).

The genuine remaining mathematics. The passage from the continuum hazard bound (Proposition 6) to the *integer* count at native resolution is **not** a write-up. Reducing the offset count by the Erdős–Turán inequality shows it is controlled by an empirical exponential sum $E_h = \frac{1}{M}\sum \mathbf{e}(h\, o_d/S)$ over the integer seeds, and that the bounded-variation bound on the density gives only an $O(1)$

discrepancy, which dominates the $O(1/S)$ main term. What is needed is genuine cancellation in E_h up to native frequency $h \sim S$: elementary on each depth-d cylinder (a linear exponential sum), but requiring that the $\asymp \lambda^d$ cylinders' resonances not conspire. This cross-cylinder estimate is the precise analogue of the characteristic-function-on-$\mathbb{Z}/3^n\mathbb{Z}$ bound that is the hardest step of Tao's theorem. It is confirmed numerically ($S \cdot \Pr[o_d \in \{1,3,5\}] \approx 3$–$4 \approx 6h(0)$, flat in d) but not proved. The companion note *Sequential density bound and the integer offset count* works this out.

What is not claimed. The density form is not asserted as an unconditional theorem: it rests on the cited spectral-gap and local-limit theorems (hypotheses discharged) *and* on the arithmetic exponential-sum estimate just described, which is open. The measure form is unconditional. The specific certified orbit $(6, 13)$ is μ-generic and deciding its own fate is Collatz-hard; nothing here bears on it. This is the same three-tier posture — unconditional generic result, conditional sharp result, untouched pointwise result — that Tao's theorem [1] occupies relative to the Collatz conjecture, and the sharp result is conditional on exactly the same kind of high-frequency arithmetic cancellation.

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